MTH 202 Chapter 5: Topics In Vector Calculus

5.5 Surface Integrals

Motivation:
In section 5.2.1 we discussed the idea that a line integral was useful for adding up a quantity over a given curve C. In this section we wish to extend this idea to adding up a quantity over a given *surface* σ to create the idea of a surface integral.

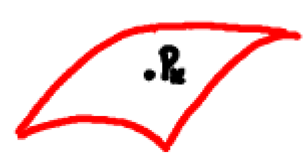
This idea can be useful in a variety of applications
- Adding up charge across a magnetized surface.
- Adding up gravitational forces across a surface.
- Adding up the density across a surface to help find the mass of the surface.
- Adding up flow of fluid through σ to help find the total quantity of fluid flowing through σ.

So, how do we construct an integral that will do this?

Exploration: Suppose we have a smooth surface σ that we plan to move across, and we want to add up the quantity at each point given by the function $f(x, y, z)$. Let Q represent the total quantity over the surface. How do we find Q?

We can begin by partition σ up into n patches.

Consider the k^{th} patch and



This leads us to the following definition.

Definition: Assuming that f is continuous, the surface integral of f over the surface σ is $\iint_\sigma f(x,y,z)\,dS$.

Remark: Let $\mathbf{r}(u,v) = x(u,v)\mathbf{i} + y(u,v)\mathbf{j} + z(u,v)\mathbf{k}$ be the parameterization of the surface σ. We saw in section 4.4 that the surface area of the k^{th} patch, ΔS_k, is approximated by $\Delta S_k \approx \|\mathbf{r}_u \times \mathbf{r}_v\| dA$.

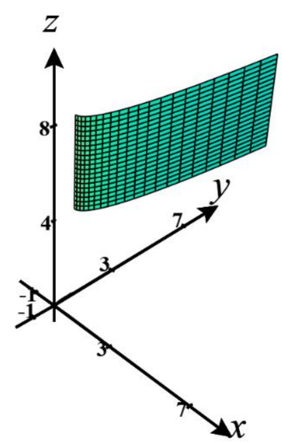
Therefore, we can rewrite the previous definition as follows.

Definition (Updated): Assuming that f is continuous, the surface integral of f over the surface σ is

Remark: Note that if $f(x,y,z) = 1$, then the surface integral $\iint_\sigma f(x,y,z)\,dS$ becomes $\iint_\sigma dS = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA$ which we know from section 4.4 yields the surface area of σ.

Example 1: Consider the surface $\sigma = \{(x,y,z) \mid y = 2x^2 + 1,\ 0 \le x \le 2,\ 4 \le z \le 8\}$.

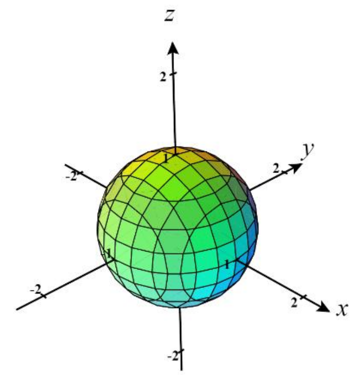
a) Find a parameterization $\mathbf{r}(u,v)$ for σ.



b) Evaluate $\iint_\sigma (xz^2)\,dS$.

Example 2: Compute the surface integral $\iint_{\sigma} z^2 dS$ where σ is the unit sphere $x^2 + y^2 + z^2 = 1$.

Hint: Use what you know about spherical coordinates to find parametric equations for σ.



<u>Example 3:</u> Suppose σ is a smooth surface given by $z = g(x, y)$ where $(x, y) \in D$. Express the surface integral $\iint_{\sigma} f(x, y, z)\, dS$ as a double integral in terms of *x* and *y*.

Note: In this example we are assuming that f is continuous on σ. Oftentimes this condition is just assumed without being stated in the textbook.

The most recent example is the justification for the following formulas.

Formulas: Suppose σ is a smooth surface given by $z = g(x, y)$ where $(x, y) \in D$. Then,

1. A normal vector to the surface at $(x, y, g(x, y))$ is $\langle -g_x(x, y), -g_y(x, y), 1 \rangle$.
2. $\iint_{\sigma} f(x, y, z)\, dS = \iint_{D} \left[f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \right] dA$

Note: It is possible to derive similar formulas in the cases where the surface is given by $y = h(x, z)$ or when the surface is given by $x = j(y, z)$. Can you guess what the formulas would be in these cases? See page 1133 of the book for one of the formulas.

Example 4: Fully setup, but do not evaluate, the surface integral $\iint\limits_{\sigma} 2\cos(y)\left(z-\tan(y)+1\right)dS$ where σ is the portion of the surface $z=\tan(y)$ which exists over the triangular region with endpoints $(0,0)$, $(1,0)$, & $(0,1)$.

We now state a fact which we will recognize as an extension of a property of line integrals.

Fact: If σ is a finite union of smooth surfaces: $\sigma_1,\sigma_2,\ldots,\sigma_n$ (we say σ is <u>piecewise-smooth</u>), then

$$\iint\limits_{\sigma} f(x,y,z)\,dS = \iint\limits_{\sigma_1} f(x,y,z)\,dS + \iint\limits_{\sigma_2} f(x,y,z)\,dS + \cdots + \iint\limits_{\sigma_n} f(x,y,z)\,dS.$$

Example 5: Suppose that σ is the "can without a top" which consists of $\sigma_1=\{(x,y,z)\,|\,x^2+y^2=1, 0\le z\le 2\}$ and $\sigma_2=\{(x,y,z)\,|\,x^2+y^2\le 1,\ z=0\}$.

a) Find a parameterization $\mathbf{r}(u,v)$ of σ_1.

b) Find a parameterization $\mathbf{s}(u,v)$ of σ_2.

c) Evaluate $\iint_{\sigma} \left(x^2 + y^2 + z^2\right) dS$.

d) Note that σ_2 didn't *need* to be parameterized and could have just been expressed as the surface in the form of $z = f\left(x, y\right)$ over a region *D*. Find the surface integral for σ_2 using this approach.